

Habib University

CS/CE 224/272 Object-Oriented Programming and Design

Methodologies

Fall 2025

Project:

- Design and implement any application/game using Object Oriented Programming (OOP) and Simple Fast Multimedia Library (SFML) 2.5.x or higher. You may use
 - SFML (<https://www.sfml-dev.org/index.php>) for game development, and
 - TGU (<https://tgui.eu/>) – (a library that is based on SFML) for application development.

For using any library other than SFML (TGUI), instructor approval is required.

- For those developing a game, you are free to use external assets for your game ***provided their sources are duly acknowledged.***
- You will be working in a team of 2 – 3 members.
- Project Teams must be from the same lecture/lab sections.

The application/game must incorporate the ***following features***:

- A welcome screen: presents startup options for the application/game.
- The application scope should be significant to justify a project (either several classes/objects interacting with each other or few classes with significant complexity or both). In a reasonably scoped project, there will be 8-10 classes.
- Background and other sounds should be added as appropriate.
- Some of the concepts learned in the course should be applied in the application/game such as inheritance, polymorphism as well as class relationships such as composition, aggregation and association.

Grade Distribution:

- ***Project Document (20%):*** You need to submit the project idea and group composition by Friday **07 November, 2025**, on the course's Canvas site.
 - (a) Include the game/application details within the proposal document. You may use game storyboarding techniques to describe the game. (refer to <https://www.cs.cornell.edu/courses/cs3152/2021sp/labs/design1/>)
 - (b) Include a UML-based design of the classes within your project.
 - (c) Include sample graphics and screens that will be used in your project.

- ***Project Progress Demo (20%):*** Starting from the week of Monday **November 17, 2025**, you have to show a progress of the project to your lab instructor during the lab session. This progress will be in the form of a partial project demo.

- ***Final Evaluation (carries 60% of overall project marks):*** Final evaluation will be done during the last week of classes i.e., starting from **Monday, December 1, 2025 through Friday, December 5, 2025**. A detailed rubric for the final evaluation is given below. As part of your final submission, you will need to submit **the updated UML specification and your final source code**.

Note: 20% penalty for submission up to 48 hours after the due date. No submissions will be graded after 48 hours.

GitHub Guidelines for Student Projects

1. **Repository Setup: Make the Repository Private:** When setting up your GitHub repository for the project, ensure that it is set to private. This will prevent unauthorized access and keep your work secure.
2. **Frequent Commits:**
Make regular commits to the repository to track your progress. Commits should be meaningful and reflect the incremental work you are doing on the project. A good practice is to commit at the end of each work session, or whenever you complete a task, to maintain a history of changes.
3. **Active Participation:** All students in the group are expected to actively participate in the project. Each member should contribute by committing their work and collaborating effectively with the team.
4. **Document Your Work:** Include clear commit messages that describe the changes you have made. Write README files and other documentation to help your team and others understand the project structure, purpose, and usage.

Project Rubric:

Proposal Document:

Component	Description	Marks
Team Formation	Team formed according to instructions	5
Game / Application Description	Is the description complete? Are there any incomplete or missing details?	5
Initial UML Diagram	UML Diagram includes all classes and their relationships	5
Screens / Panels	Are the screens representative of the game or application? Do the screens describe main parts of the game?	5
TOTAL		20

Project Progress Demo:

Component	Description	Marks
Project Demo	A running project demo is shown, with no errors	10
Percentage of Work	The project demo shows at least 50% of the features mentioned in the proposal	5
Questions/ Answers	Team Members are able to answer questions	5
TOTAL		20

Final Project Demo Rubric:

Component	Description	Marks
Project Demo	A running project demo is shown, with no errors	5
Percentage of Work/Scope	The project demo includes all of the features mentioned in the proposal.	10
Questions/ Answers	Team Members are able to answer questions	10
Code	The code has no warnings (3), is properly commented (3), is modular (3), has proper indentation (3) is easy to understand (3), and is submitted on time (3).	10
Documentation	Proper documentation in the form of README or other format(s) is included.	10
Object Orientation	The code includes core object oriented features such as inheritance and polymorphism with good	15

	effect.	
TOTAL		60